Data Collection: Part 2 (Unstructured)

Data is everywhere
Access takes effort
Structure is earned

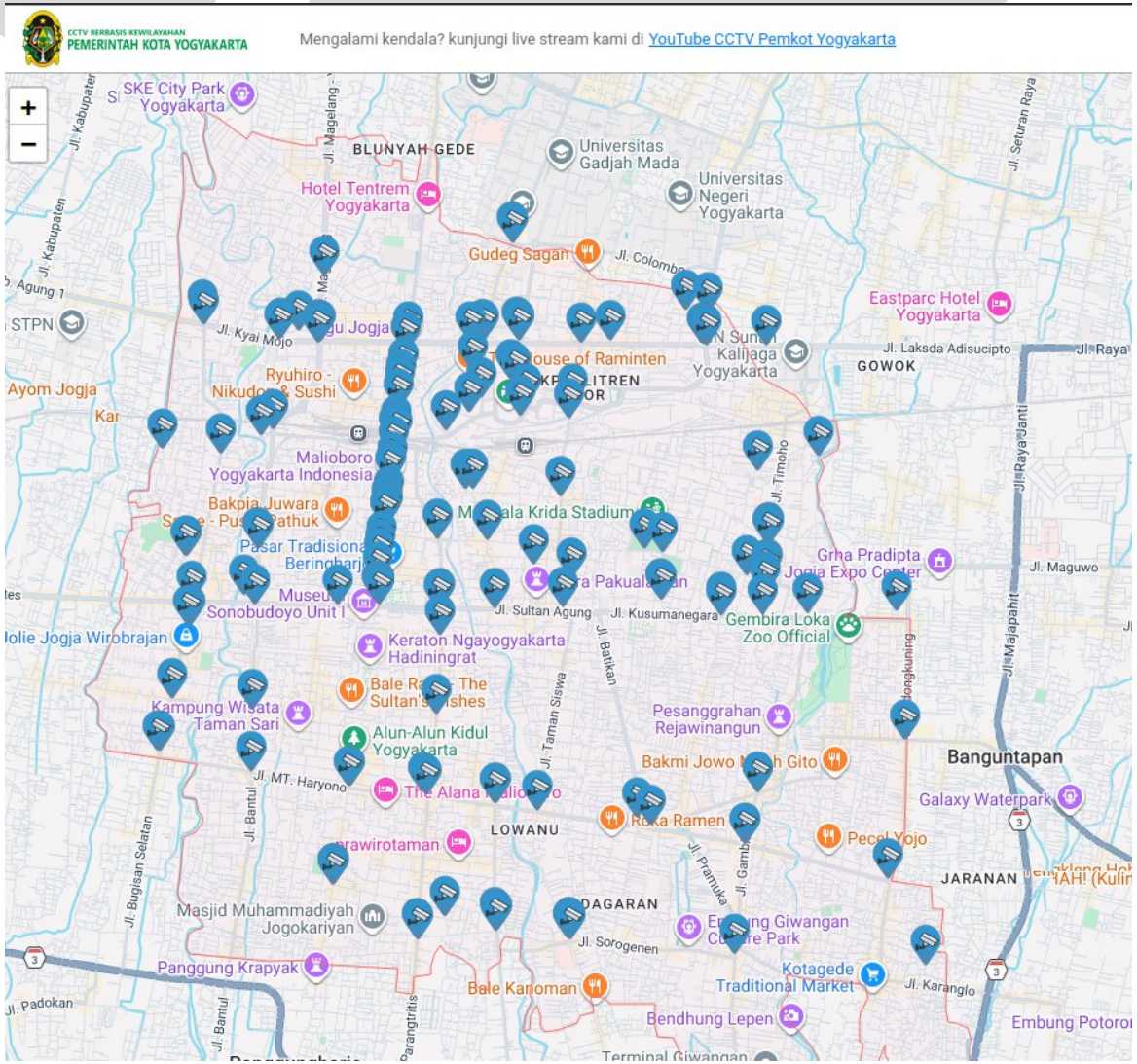
https://github.com/jendralhxr/introdatascience

Semi-structured and Unstructured Data

The way data are organized varies a lot:

- Format heterogeneity
- Access limitation and **license** (both legally and morally)
- How to infer information (extracting features) is still another step

Live Traffic Video: Jogja





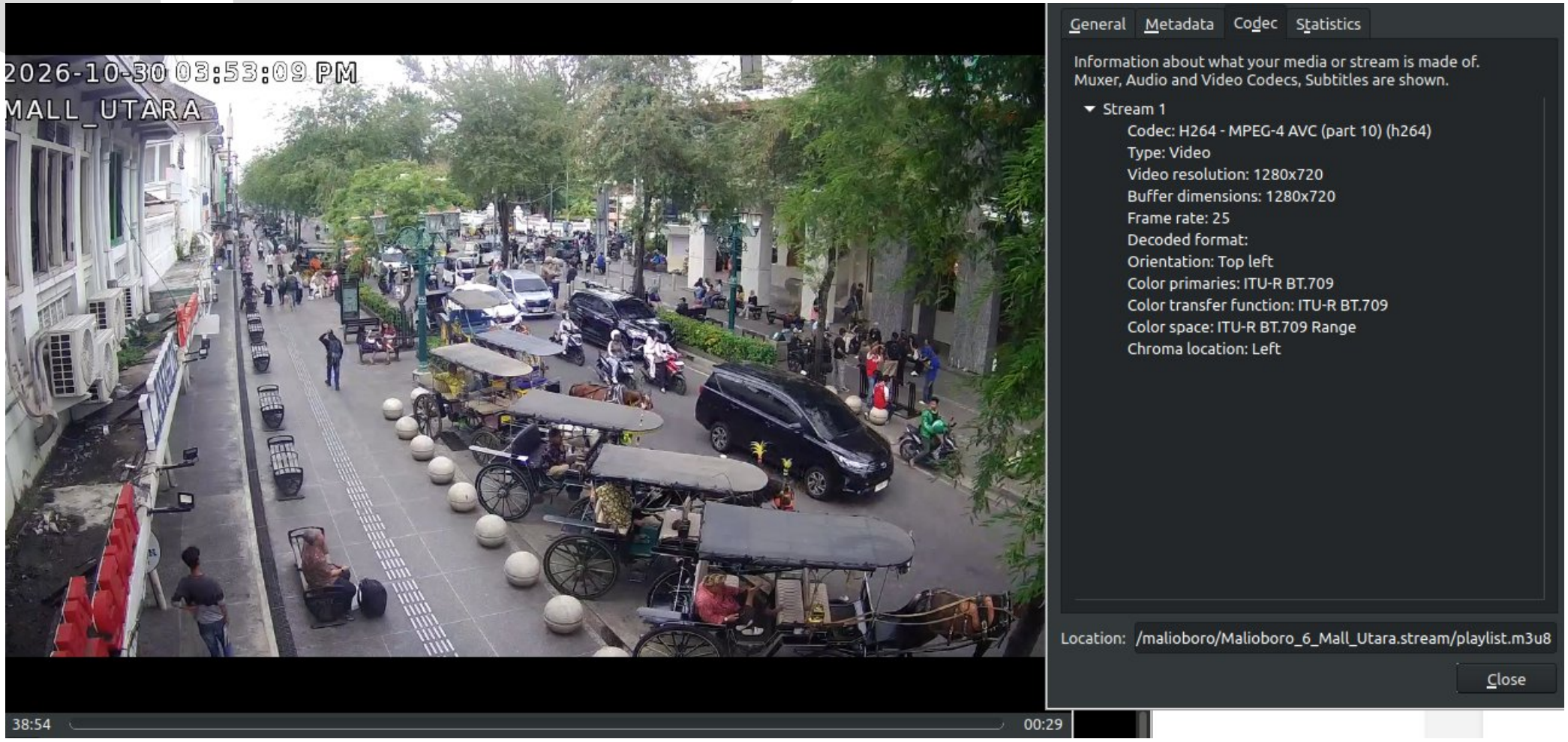
https://cctv.jogjakota.go.id/

Attendance List



https://forms.gle/q7yN33EmTYT5dpLB6

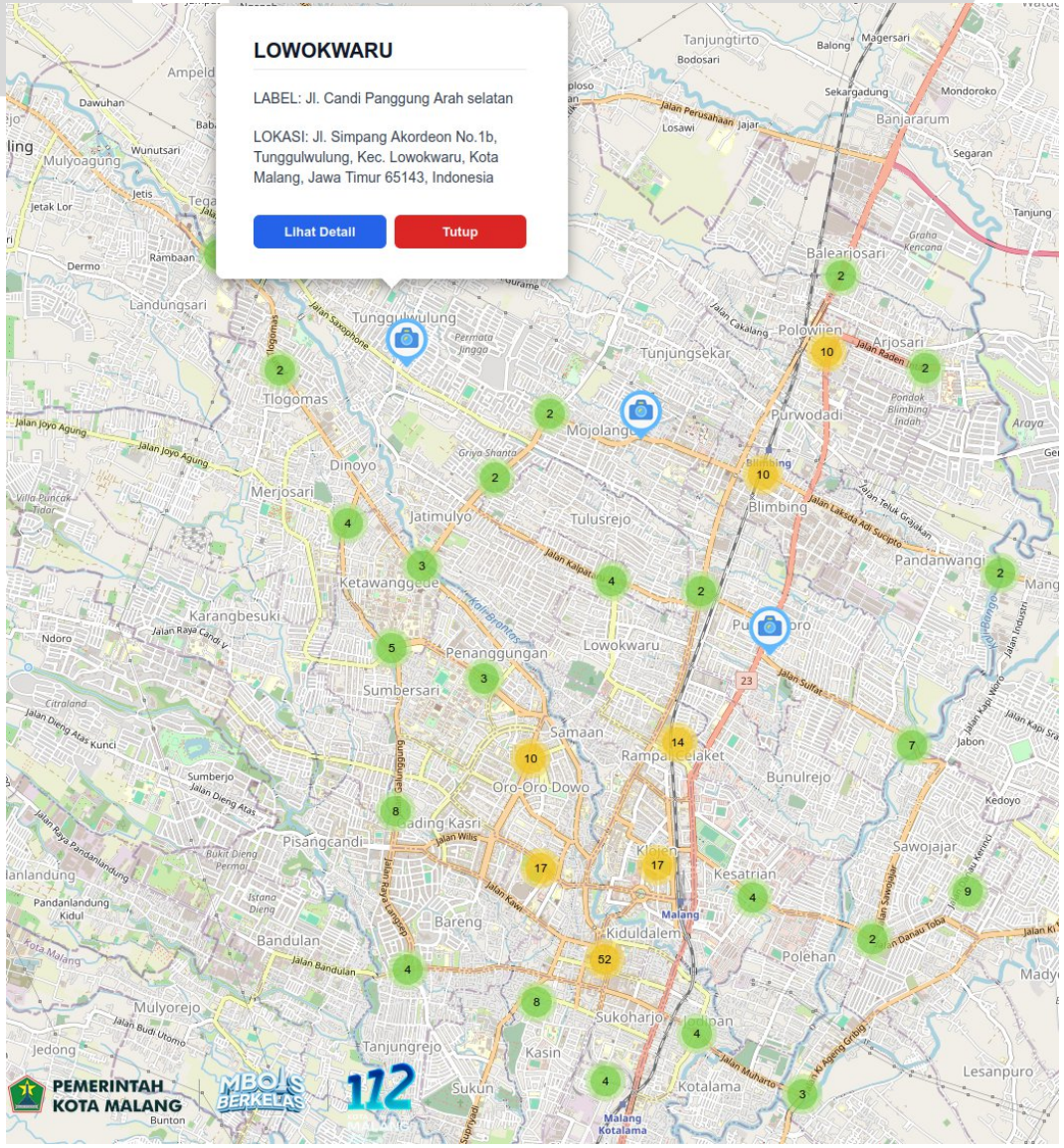
Live feed: allowed to access

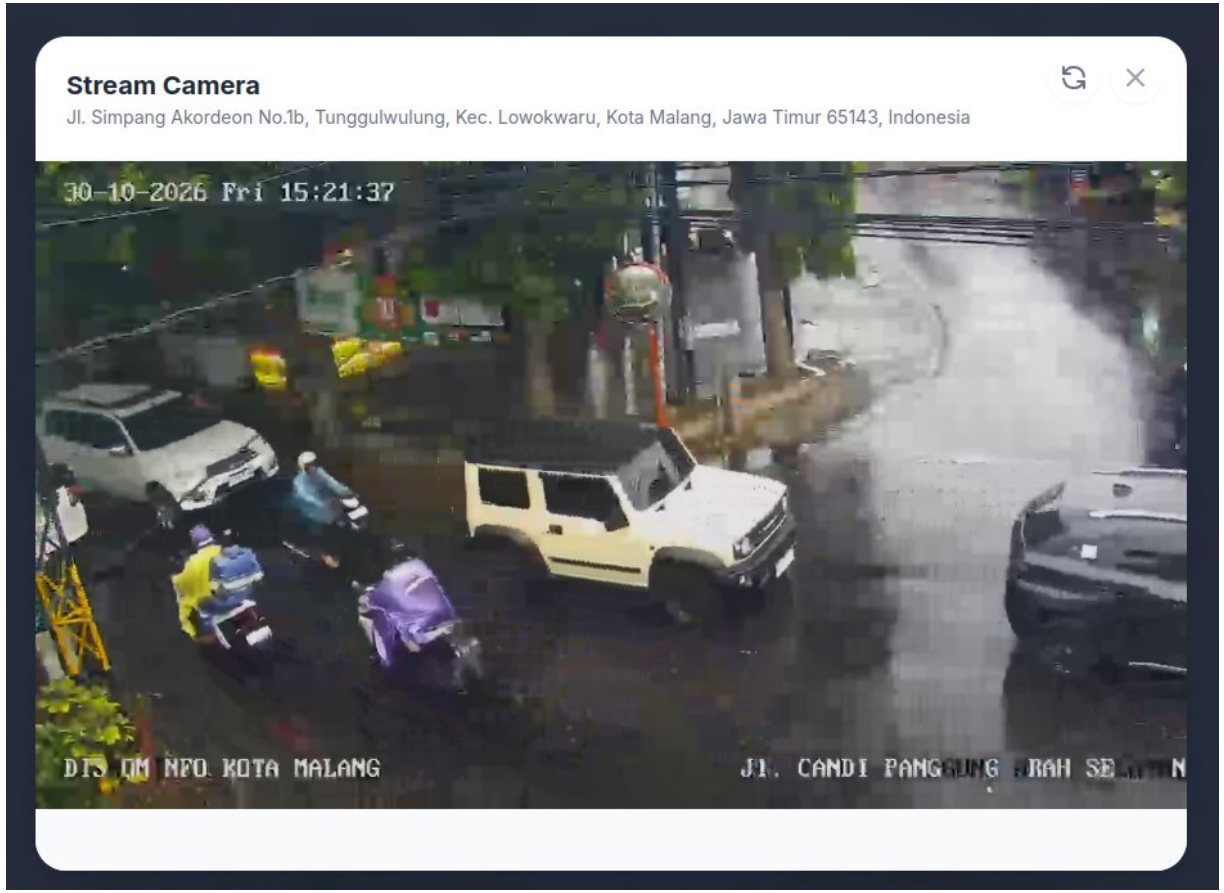


vlc

`https://cctvjss.jogjakota.go.id/malioboro/Malioboro_6_Mall_Utara.stream/playlist.m3u8`

Live Traffic Video: Malang



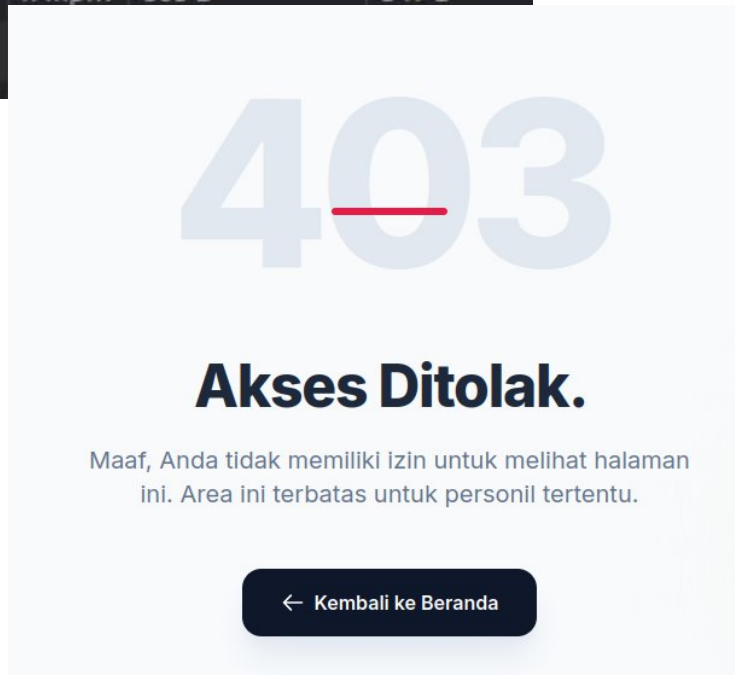


https://cctv.malangkota.go.id/sebaran-cctv

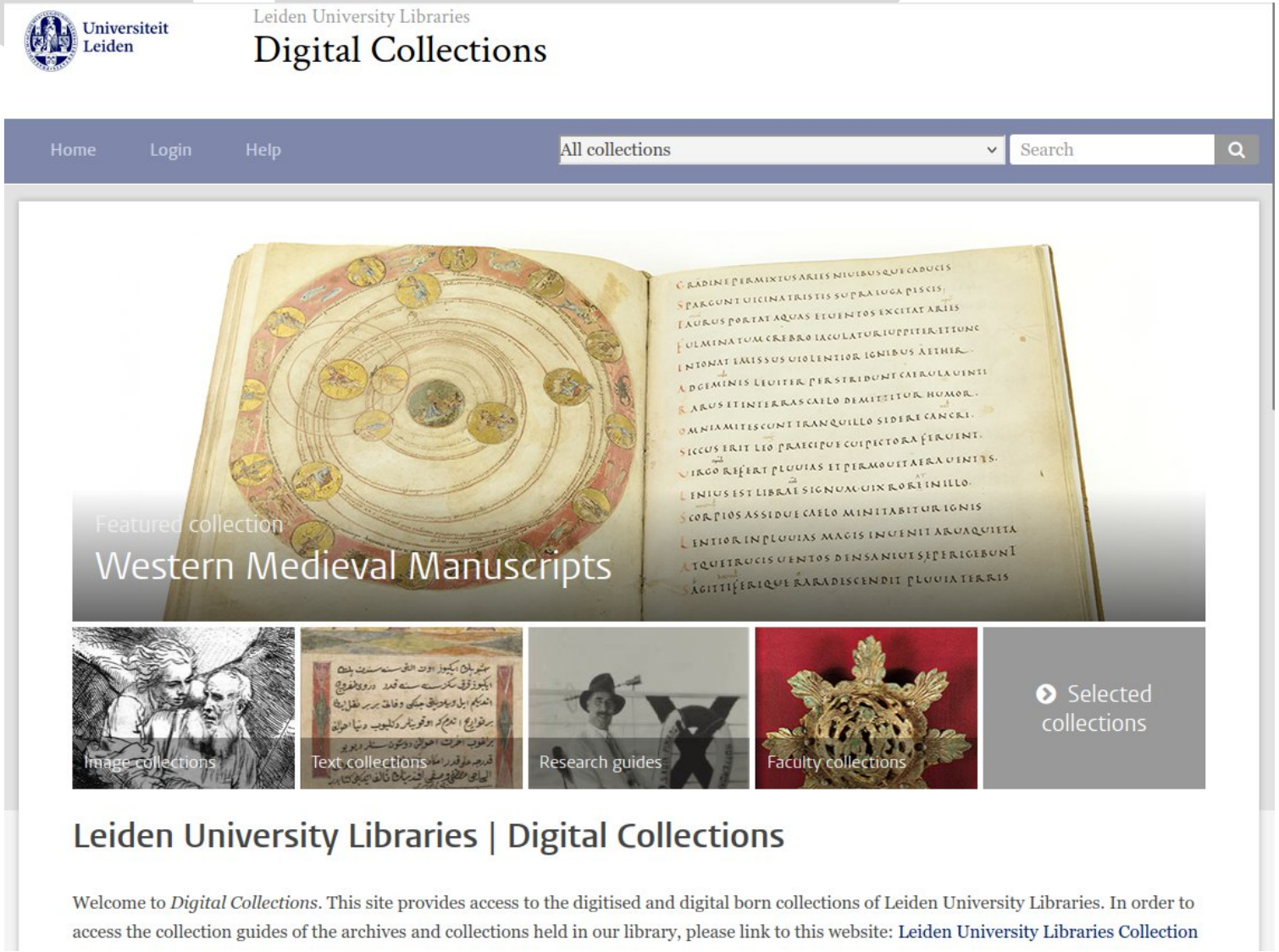
Video frame sequence: not allowed

200	GET	cctv.malangkota.go.id	807179387709306202506877.m3u8	hls.js@latest:3 (xhr)	x-mp…	865 B	347 B
200	GET	cctv.malangkota.go.id	807179387709306202506877_0p87433.ts	hls.js@latest:3 (xhr)	mp2t	86.16 kB	85.73 kB
200	GET	cctv.malangkota.go.id	807179387709306202506877.m3u8	hls.js@latest:3 (xhr)	x-mp…	865 B	347 B
200	GET	cctv.malangkota.go.id	807179387709306202506877_0p87434.ts	hls.js@latest:3 (xhr)	mp2t	56.64 kB	56.21 kB
200	GET	cctv.malangkota.go.id	807179387709306202506877.m3u8	hls.js@latest:3 (xhr)	x-mp…	865 B	347 B
200	GET	cctv.malangkota.go.id	807179387709306202506877_0p87435.ts	hls.js@latest:3 (xhr)	mp2t	67.17 kB	66.74 kB
200	GET	cctv.malangkota.go.id	807179387709306202506877.m3u8	hls.js@latest:3 (xhr)	x-mp…	865 B	347 B
200	GET	cctv.malangkota.go.id	807179387709306202506877_0p87436.ts	hls.js@latest:3 (xhr)			

```
streamlink https://cctv.malangkota.go.id/cctv-stream/streams/807179387709306202506877.m3u8
[cli][info] Found matching plugin hls for URL https://cctv.malangkota.go.id/cctv-stream/streams/807179387709306202506877.m3u8
error: Unable to open URL: https://cctv.malangkota.go.id/cctv-stream/streams/807179387709306202506877.m3u8 (403 Client Error: Forbidden for url: https://cctv.malangkota.go.id/cctv-stream/streams/807179387709306202506877.m3u8)
```



Medieval/Ancient Document Library



(Sino-)Malay popular fiction (KITLV)
Archive Inventories
Archives and Letters
Boekenoogen Chapbooks (MNL)
Caribbean Maps
Caribbean Sources
Colonial Archives (KITLV)
Colonial Journals (KIT)
Colonial Sources (KIT)
Drawings (Print Room)
Dutch Colonial Maps KIT
Dutch East Indies literature (KITLV)
Faculty collections
Frank Scholten Photographs, Negatives and Photo Albums
Image collections
Indonesian Printed Resources
Indonesian Serials (NIOD)
Indonesian Special Collections
Leiden Icons
Library Postcards Sjoerd Koopman
Manuscripts, Archives & Letters
Maps & Atlases
Maps (KITLV)
Naxi Dongba manuscripts
North Korean Posters
Online Exhibitions
Oral History Archive CIHC
Oral History Archive SMGI (KITLV)
Photo Albums
Photographica (Print Room)
Photography (Kern Institute)
Photography (Print Room)
Prints (Print Room)
Rare Books
Research guides
Selected collections
Southeast Asian & Caribbean Images (KITLV)
Southeast Asian Pop Music (KITLV)
Text collections
Tibetan Texts (Kern Institute)
Topographic prints and drawings
UBL Library Catalogues
Unofficial Poetry from China
Van Gulik and Chinese Culture
Western Medieval Manuscripts

A collection

Or. 1990

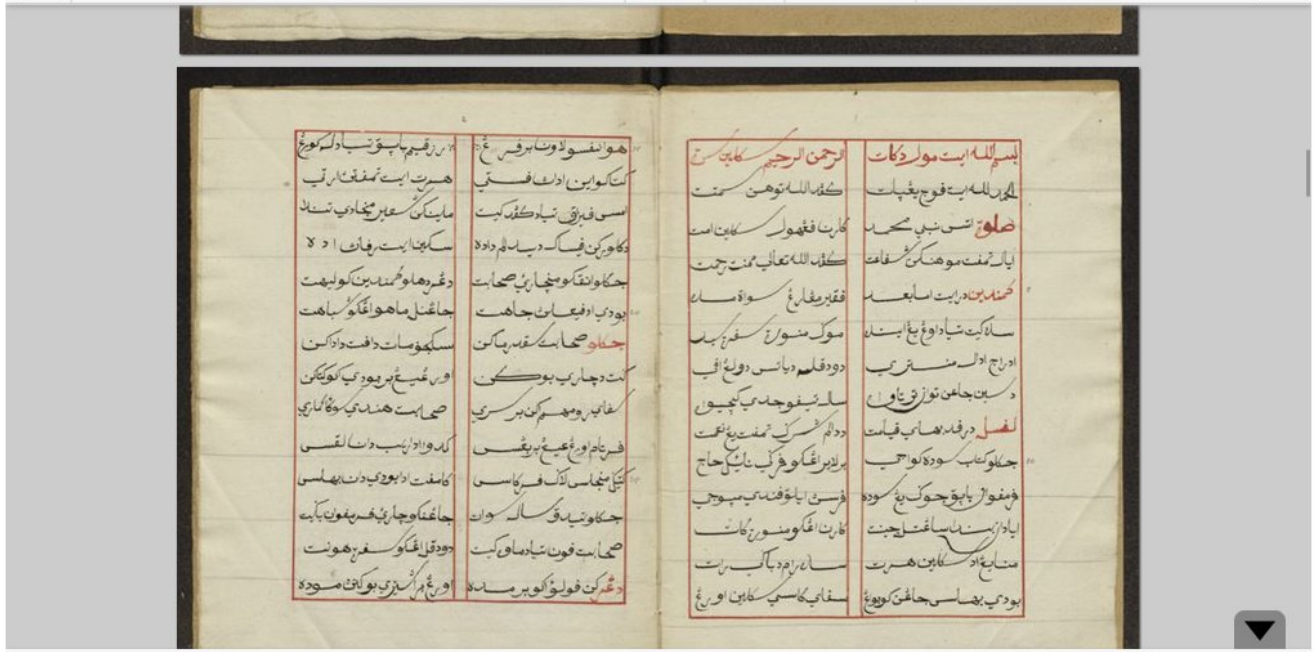
View pages

View | Pages

Or. 1990...

TEXT

Search inside this book



Page 3 of 13

Description

This object has been migrated. Please click here to view the object in our new system.

Persistent URL	http://hdl.handle.net/1887.1/item:1934486
Title	Syair Ikan
Creator/other	Muhammad Hasan. Batten.
Shelfmark	Or. 1990

Advanced Viewer

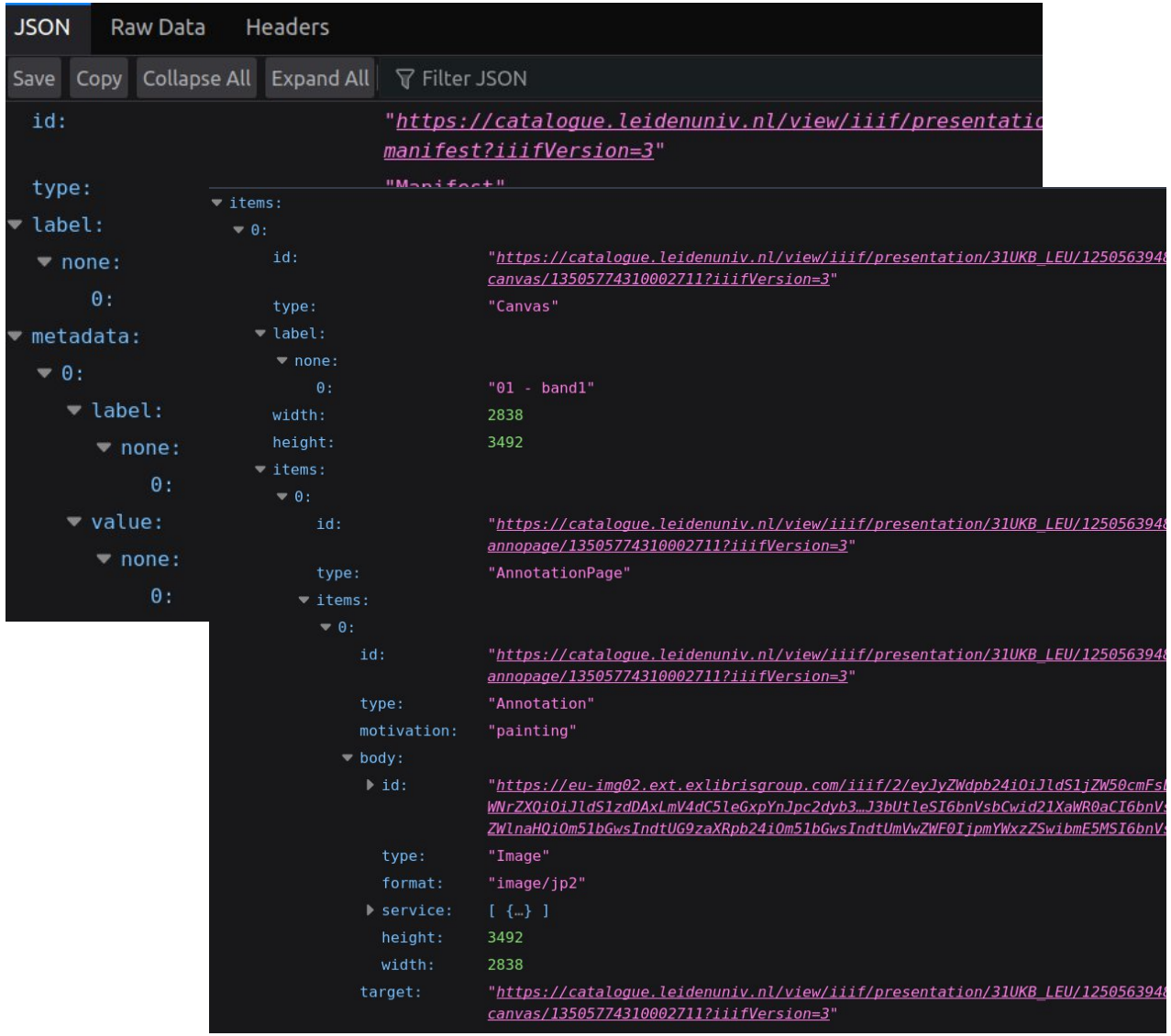
Digital Collections

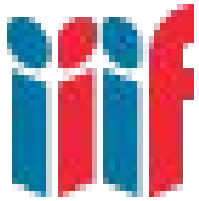
This item can be found in the following digital collections:

Manuscripts, Archives & Letters

manifest.json: semi-structured list of assets

Persistent URL	http://hdl.handle.net/1887.1/item:1934486
Title	Syair Ikan
Creator/other	Muhammad Hasan. Batten.
Shelfmark	Or. 1990
Other version identifier	81321322780002711
Language	Malay
Country	Netherlands
Form	13 pages
Extent	13 scans
Published/created	Jakarta : Batten, 1865
Restriction on access	Full access.
Use and reproduction	The rights status of this resource is public domain. Citing the Leiden University Libraries as a source is appreciated. PUBLIC DOMAIN
IIIF manifest	iiif manifest





IIIF

IIF is a set of open standards for delivering high-quality, attributed digital objects online at scale. It’s also an international community developing and implementing the IIIF APIs. IIIF is backed by a consortium of leading cultural institutions.

Downloading the images

https://github.com/jendralhxr/whatevering/blob/master/iiifmanifestdownload.py

```python
# Get the IIIF image service ID or base ID
service = body.get("service", [])
if service:
    service_id = service[0].get("id")
    # Construct IIIF full-resolution URL
    img_url = f"{service_id}/full/full/0/default.jpg"
else:
    img_url = body.get("id")

# Clean label for file path
safe_label = "".join(
    c if c.isalnum() or c in (" ", "-", "_") else "_" for c in label
)
file_name = f"{idx:02d}_{safe_label}.jpg"
file_path = os.path.join(output_dir, file_name)

print(f"[{idx}/{len(canvases)}] Downloading {file_name}...")

# Download full resolution image
req = urllib.request.Request(
    img_url, headers={"User-Agent": "Mozilla/5.0"}
)
with urllib.request.urlopen(
    req, context=ssl_context
) as response, open(file_path, "wb") as f_out:
    f_out.write(response.read())
```

```
Found 13 pages in manifest. Starting download...

[1/13] Downloading 01_01 - band1.jpg...
[2/13] Downloading 02_02 - band2.jpg...
[3/13] Downloading 03_03 - opening01.jpg...
[4/13] Downloading 04_04 - p001-002.jpg...
[5/13] Downloading 05_05 - p003-004.jpg...
[6/13] Downloading 06_06 - p005-006.jpg...
[7/13] Downloading 07_07 - p007-008.jpg...
[8/13] Downloading 08_08 - p009-010.jpg...
[9/13] Downloading 09_09 - p011-012.jpg...
[10/13] Downloading 10_10 - p013-014.jpg...
[11/13] Downloading 11_11 - p015-016.jpg...
[12/13] Downloading 12_12 - p017-018.jpg...
[13/13] Downloading 13_13 - opening02.jpg...

All downloads finished!
```



بسم الله ایت مولد کات
الحمد لله ایت فوج یغ پات
صلوة اسی نبي محمد
ایاله تمفت موهنکن شفاعت
کمندین درایت امابعد
ساله کیت تیادا وغ یغ اینده
ادراج اداله منتریي
دسین جاعن توان تریاوا
لفسله درفد بهاي قیامت
جکلو کتاب سوده کوا جب
فرمفوان بابو جوک یغ سوده

الرحمن الرحیم سکاین سی
کفد الله توهن سمت
کارن فغهول سکاین امت
کفد الله تعالي ممنت رحمت
فقیر مقارغ سواة ماله
موک منورة سفرن کید
دود قلم دیاسی دولغ افي
ساله تیفو جدي کچیوا
ددالم شرکن تمفت یغ نعمت
برلابر اغکو فرکي نایک حاج
فرسن ایاو فندي مپوجب
کارن اغکو منورة کات
ساله رام دباکي رات
سفاي کاسمي سکاین اورغ

هوا نفسو لاون برفرغ
کتاکو این اداث فستي
امسی فیرق تیاد کفد کیت
دکلو رکن فیساک دیدالم داده
جکلو انفکو منجاریي صحابت
دغر دهلو کمندین کولیهت
بودي ادفیعلن جاهت
جکلو صحابت کفد رماکن
کنت دجاري بوکن
کفاي رومهم کن برسري
فرتام اورغ عیغ بربقسی
کیکي منجلسی لاک فرکاسی
جکلو تیدق ساله کوات
صحابت فون تیادماو کیت
دغر کن فولوکو برمده

ر رقیم بابو تیادله کورغ
هرت ایت تمفت ارتب
ماینکن کعین منجادي تنله
سکین ایت رفاث اده
دغر دهلو کمندین کولیهت
جاغنل ماهو اغکو کباهت
سکجومات دافت داداکن
اورغ عیغ بربودي اکو کتاکن
صحابت هندي کوکاماري
کدوا اداربب دان لقسی
کامفت ادابودي دان بهاسی
جاغنکو چاري فرمفوان بکیت
دودقل اغکو کفرهونت
اورغ برکتري بوکتن موده

Descriptive Metadata (MODS) 242 bytes
Descriptive Metadata (DC) 448 bytes
Original master 55.9 MB
Thumbnail 25.5 KB
Image (medium resolution) 66.6 KB
Image (high resolution) 1.2 MB
Technical metadata 9.1 KB

https://digitalcollections.universiteitleiden.nl/view/item/1934488

Formats





Descriptive Metadata (MODS)	242 bytes
Descriptive Metadata (DC)	448 bytes
Original master	55.9 MB
Thumbnail	25.5 KB
Image (medium resolution)	66.6 KB
Image (high resolution)	1.2 MB
Technical metadata	9.1 KB

Size in pixels: 5512 × 3541 pixels
Print size: 1944.5 × 1249.2 millimeters
Resolution: 72 × 72 ppi
Color space: RGB color
Precision: 8-bit gamma integer

File Name: /me/pro/curs...001-002.jpg
File Size: 2.5 MB
File Type: JPEG image

JPEG

TIFF

Name:
OBJ Datastream.tiff
Width:
5512 pixels
Height:
3541 pixels
Type:
TIFF image
File size:
58.6 MB